



3D topology optimization and additive manufacturability of two-fluid heat exchangers

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ABSTRACT

This paper presents a topology optimization approach aiming at enhancing the performance of two-fluid heat exchangers. The study focuses on optimizing three types of material distributions: the hot fluid, the cold fluid, and the solid material, with the objective of maximizing heat transfer efficiency. A formulation using two density fields is developed to optimize the distribution of the three materials. Two manufacturability constraints, the thickness of the solid layer and the self-supported overhang angle, are imposed to ensure practicality and feasibility. Controlling the solid layer thickness is crucial for ensuring that the thinnest wall can be fabricated using additive manufacturing. Overhang angle control is important for creating self-supported solid components that can be printed without support structures. A numerical example compares the performance of the optimized heat exchanger with a benchmark straight channel design, demonstrating a remarkable 2.3 times increase in heat transfer rate for the optimized configuration. The topologically optimized design improves the heat transfer rate by disrupting the thermal boundary layer and enhancing heat convection. Even when compared to the straight channel heat exchanger with a 1.9 times higher surface area, the optimized design still demonstrates a 12% higher heat transfer rate and achieves a power density 5.3 times higher than that of the straight channel heat exchanger. An additive manufacturing process is employed to validate the manufacturability of the optimized design, resulting in the successful fabrication of heat exchangers in three different sizes. This study underscores the effectiveness of topology optimization with manufacturability constraints in producing heat exchangers that are manufacturable and have superior heat transfer performance.

1. Introduction

Heat exchangers are devices that are used in a variety of industries and applications for the purpose of transferring heat from one fluid to another. These devices are designed to transfer energy between two fluids, a hot fluid and a cold fluid, while keeping the two fluids separated from each other by a solid layer.

Traditionally, heat exchangers are manufactured using a variety of methods, including welding, brazing, forming, and casting. Chu et al. [1] studied printed circuit heat exchangers (PCHEs) with straight channels and found that supercritical CO₂ exhibited better heat transfer capabilities than water. Liu et al. [2] examined the thermal-hydraulic model for PCHEs. Park et al. [3] presented experimental data on straight-channel PCHEs using supercritical CO₂ and proposed a discretization method for heat exchanger design.

As additive manufacturing gains popularity, engineers are increasingly exploring and fabricating heat exchangers using this technology.

Zhang et al. [4] manufactured and tested the manifold-microchannel heat exchanger for high temperature applications. They achieved a 25% improvement in heat transfer density compared to conventional plate fin heat exchangers. Sabau et al. [5] studied the tube and shell heat exchangers, achieving an overall heat transfer rate 32% higher than that of a baseline design. Dixit et al. [6] used triply periodic minimal surfaces (TPMS) structures to design heat exchangers. Experimental results demonstrated a 55% increase in effectiveness compared to a thermodynamically equivalent, most-efficient counter-flow heat exchanger, while occupying only one-tenth of the space. For an overview of related work, we refer to Kaur and Singh [7].

The aim of this paper is to use topology optimization (TO) to exploit the geometric freedom of additive manufacturing, thereby enhancing heat transfer performance in heat exchanger design.

Topology optimization is a design approach originally used to study the distribution of materials to optimize structural performance (Bendsoe [8]). It was used to study the optimization of Stokes flow by

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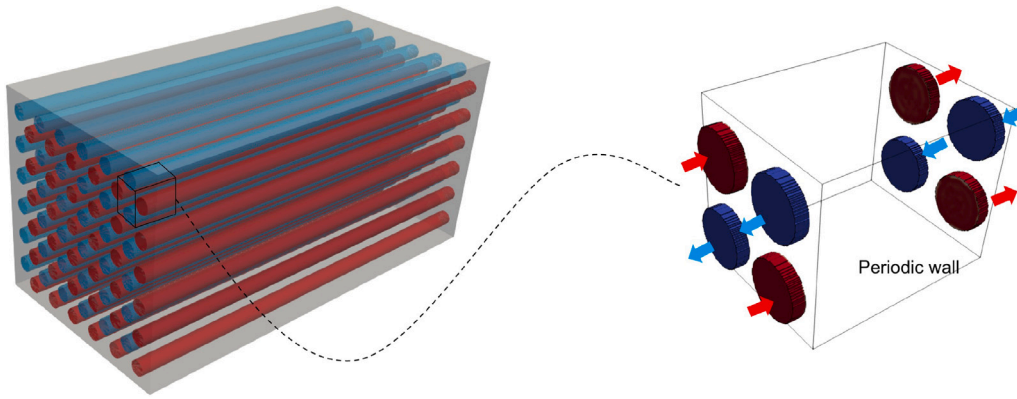


Fig. 1. Heat exchanger core with a two-fluid system. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Borrval and Petersson [9]. For the conjugate heat transfer problem, Yaji et al. [10] studied 3D designs using a level-set method. Zeng et al. [11] used a multiple-layer method to design a heat sink for forced air cooling. Sun et al. [12] investigated the topology optimization of manufacturable heat sinks, which were additively manufactured and experimentally tested to compare with the CFD predictions. Alexandersen et al. [13] tackled the natural convection problem with up to 330 million state degrees of freedom. Their optimized heat sinks were experimentally studied in Lazarov et al. [14].

Unlike heat sinks, heat exchangers are two-fluid systems, where the two fluids must be separated by a solid layer. Kobayashi et al. [15] optimized the fin and tube heat exchangers with a solid plate between the two fluids. Høghøj et al. [16] optimized a two-fluid system, generating the solid layer through an erosion-dilation process on a single density field. Kobayashi et al. [17] used a filter to create a region between the two fluids. Feppon et al. [18] used minimal thickness control in the level set method. Galanos et al. [19] used the one density method from Høghøj et al. [16] and conducted a successive shape optimization following the topology optimization of the heat exchangers. The Spalart–Allmaras Model was used in their study.

This paper optimizes a three-material two-fluid heat exchanger with two density fields. In previous works such as those by Kobayashi et al. [17] and Høghøj et al. [16], one density field was used to represent three materials. The density value representing the solid was sandwiched between values representing the two fluids. This approach results in the generation of a solid layer between the two fluids. In our study, however, two density fields are utilized to represent the three materials, allowing the free exchange of three materials. The two fluids are separated with a layer of solid wall, whose thickness is controlled by the minimal wall thickness constraint imposed by the additive manufacturing process. Our work is, to the best of our knowledge, among the first to consider manufacturability in the topology optimization of a two-fluid heat exchanger. While previous studies have explored the topology optimization of heat exchangers, they have either not fully addressed manufacturability or have focused on single-fluid models. The two manufacturing constraints considered in this paper are the minimal wall thickness and the overhang angle. The minimal wall thickness is simply controlled by a numerical value as stated above. The overhang angle, on the other hand, is controlled by density-gradient based overhang angle constraint (Qian [20]).

The paper is organized as follows. Section 2 describes the general design problem for two-fluid heat exchangers. Section 3 describes the topology optimization method used in this work. Section 4 presents a discussion on the optimized heat exchanger and its performance, followed by Section 5, where we showcase the additive manufacturing of the heat exchangers in three different sizes along with X-ray scans. Finally, the conclusions are drawn and future work is discussed in Section 6.

2. Problem description

The purpose of this study is to explore new channel geometries to enhance the inter-fluid heat transfer rate while keeping a low pressure drop. In this section, we discuss the general design problem for the two-fluid system. Fig. 1 (left) shows the heat exchanger core with straight channels. The red color represents the hot fluid channels, the blue color represents the cold fluid channels, and between them are the solid components in gray. To achieve a higher heat transfer rate, a counter flow is usually used, which means the hot fluid enters from one side of the core and exits the other, with the cold fluid flowing in the opposite way with respect to each adjacent channel. Fig. 1 (right) depicts the counter flow design. The channels have circular cross sections to reduce pressure drop. We modeled the fluid flow using Navier–Stokes (NS) equations and solved the temperature distribution using the energy equation.

However, directly using CFD to model the entire core is computationally expensive. Especially in topology optimization which needs to solve the NS and the adjoint equations hundreds of times. We thus choose to design a segment of the full core and leverage periodicity to achieve the full scale model. As shown in Fig. 1 (right), a unit cell with two hot, and two cold fluid inlets/outlets is used as the geometry of the design problem. A periodic wall boundary condition is used on the four sides of the unit cell to ensure the optimized structure can be duplicated to fill the whole space of the full heat exchanger core. Additionally, we used flow uniformity constraints to ensure flow periodicity which will be described further in Section 3.4.

3. Topology optimization of heat exchanger accounting for additive manufacturability

In this section, we delve into the topology optimization method for heat exchangers, with an emphasis on manufacturability.

3.1. Two-density design parameterization of three-material heat exchangers

We use two density fields to represent three materials, the hot fluid, the cold fluid, and the solid component. Fig. 2 shows how the method works. The first field γ_1 is to distinguish whether it is a hot fluid or not, and the second field γ_2 is to distinguish whether it is a cold fluid or not. Thus, if $\gamma_1 = 1$, it is regarded as a hot fluid, and if $\gamma_1 = 0$, it is either a cold fluid or a solid material. Similarly, for $\gamma_2 = 1$, the field is regarded as a cold fluid, and if $\gamma_2 = 0$, the field is either a hot fluid or a solid material. By combining the two fields, we can express the three materials as shown in Eq. (1).

$$\begin{cases} \text{Hot fluid:} & \gamma_1 = 1, \gamma_2 = 0, \\ \text{Cold fluid:} & \gamma_1 = 0, \gamma_2 = 1, \\ \text{Solid:} & \gamma_1 = 0, \gamma_2 = 0. \end{cases} \quad (1)$$

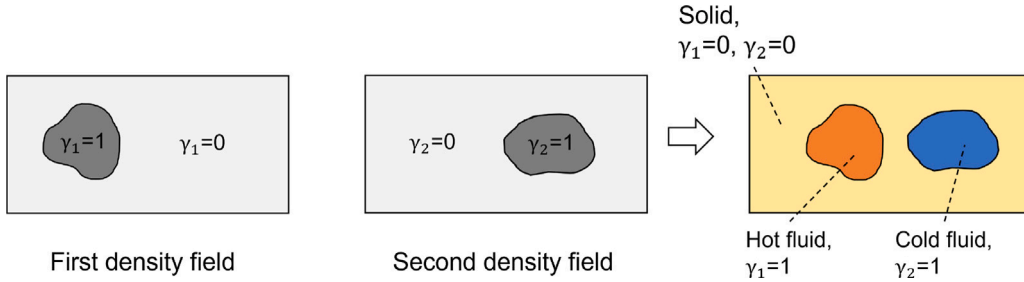


Fig. 2. Three materials represented by two density fields.

In practice, the two density fields will be filtered by a Helmholtz PDE filter (Lazarov and Sigmund [21]; Kawamoto et al. [22]) and a Heaviside projection (Guest et al. [23]) to create the physical fields that are used for material parameterization. In this way, the two optimization fields, γ_1 and γ_2 , are filtered to create the physical fields, $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$.

The equation of the Helmholtz PDE filter is given as:

$$-R^2 \nabla^2 \tilde{\gamma}_j + \tilde{\gamma}_j = \gamma_j, \quad j = 1, 2 \quad (2)$$

where R is used to control the kernel size. γ_j ($j = 1, 2$) is the optimization density fields and $\tilde{\gamma}_j$ ($j = 1, 2$) is the filtered density fields. According to Lazarov and Sigmund [21], a relationship between the kernel size R and filter radius r is given as $r = \frac{R}{2\sqrt{3}}$.

The filtered density fields from the Helmholtz filter are smoothed using the optimization fields. The physical fields are created by using the Heaviside filter.

$$\tilde{\gamma}_j = \begin{cases} \eta(e^{-\beta(1-\tilde{\gamma}_j/\eta)} - (1-\tilde{\gamma}_j/\eta)e^\beta), & \text{if } \tilde{\gamma}_j \leq \eta, \\ (1-\eta)(1 - e^{-\beta(\tilde{\gamma}_j-\eta)/(1-\eta)} + (\tilde{\gamma}_j - \eta)/(1-\eta)e^{-\beta}) + \eta, & \text{if } \tilde{\gamma}_j > \eta, \end{cases} \quad (3)$$

where $\tilde{\gamma}_j$ are the physical density fields. $\eta = 0.5$. β in Eq. (3) is initially set to 1 and doubled each 40 iterations of the optimization process while $\beta \leq 16$.

Compared with other multi-material optimization approaches (Zuo and Saitou [24], Li and Kim [25]), the two-fluid heat exchanger optimization has an additional constraint: the two fluids must not mix. Therefore, a solid layer should be formed between the hot fluid and the cold fluid during the optimization process. A minimal wall thickness constraint is introduced to achieve this goal. As shown in Fig. 3, there is an overlap region of the hot fluid and the cold fluid when both $\gamma_1 = 1$ and $\gamma_2 = 1$. This overlap region suggests that the two fluids mix together. In order to prevent the mixing, the minimal wall thickness constraint is introduced as:

$$\frac{\int_{\Omega} \tilde{\gamma}_1 \tilde{\gamma}_2 d\Omega}{V} \leq \Phi_o, \quad (4)$$

where Ω is the design domain, V is the volume of the design domain, and Φ_o is the ratio of the overlap region over the entire domain. When Φ_o equals to 0, there is no overlap region, i.e. no cold fluid and hot fluid mixing. In practice, Φ_o is chosen to be a small value in order to prevent the mixing of the two fluids.

Eq. (4) effectively prevents the two fluids from overlapping. However, the two fluids can still come into contact with each other under this constraint. In such cases, there is no solid wall separating the two fluids, resulting in a wall thickness of 0. To ensure a genuine separation between the two fluids, we replace the fields $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$ with their dilation fields. The dilation fields are created by using the Heaviside projection in Eq. (3) with a different value of η :

$$\tilde{\gamma}_j^d = \begin{cases} \eta_d(e^{-\beta(1-\tilde{\gamma}_j/\eta_d)} - (1-\tilde{\gamma}_j/\eta_d)e^\beta), & \text{if } \tilde{\gamma}_j \leq \eta_d, \\ (1-\eta_d)(1 - e^{-\beta(\tilde{\gamma}_j-\eta_d)/(1-\eta_d)} + (\tilde{\gamma}_j - \eta_d)/(1-\eta_d)e^{-\beta}) + \eta_d, & \text{if } \tilde{\gamma}_j > \eta_d, \end{cases} \quad (5)$$

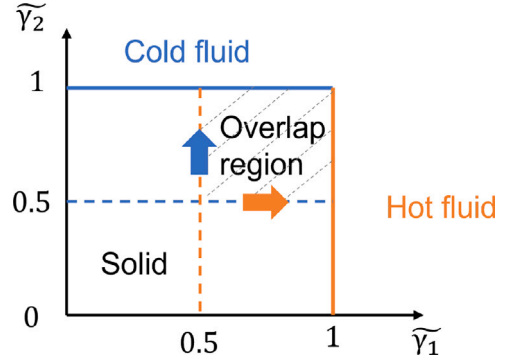


Fig. 3. Two fluids overlap with each other.

where $\tilde{\gamma}_j^d$ are the corresponding dilation fields of $\tilde{\gamma}_1$ and $\tilde{\gamma}_2$. Since $\tilde{\gamma}_j^d$ are regarded as the dilation fields of the $\tilde{\gamma}_j$, it is required that $\eta > \eta_d$. When $\eta < \eta_d$, $\tilde{\gamma}_j^d$ will be the erosion fields.

Eq. (4) is replaced with the dilation fields as:

$$\frac{\int_{\Omega} \tilde{\gamma}_1^d \tilde{\gamma}_2^d d\Omega}{V} \leq \Phi_o. \quad (6)$$

Fig. 4 illustrates the idea of generating a solid layer between the two fluids by imposing the minimal wall thickness constraint. When the yellow overlap region area becomes zero, a wall is formed between the two fluids.

3.2. Manufacturability constraints

Two types of manufacturing constraints are considered in this paper: the minimal wall thickness constraint, and the self-supported overhang angle constraint. The minimal wall thickness is the thinnest wall that the machinery can print while maintaining structural integrity, and a sealing barrier preventing the fluids from leaking into adjacent channels. Meeting the overhang angle requirement ensures that the structure is self-supported during the printing process, eliminating the need for support structures. In this work, the minimal wall thickness is 0.4 mm and the maximum overhang angle is 45°.

3.2.1. Minimal thickness of the solid wall

The minimal thickness of the solid layer is influenced by the minimal thickness of the dilation fields. Specifically, two factors contribute to controlling this minimal thickness in the PDE filter and the Heaviside projection: filter radius r , and the difference between η and η_o . Fig. 5a shows the physical field $\tilde{\gamma}$ and its dilation field $\tilde{\gamma}^d$ by selecting $\eta = 0.5$, and $\eta_o = 0.05$. Since $\tilde{\gamma} = 1$ is recognized as fluid, $\tilde{\gamma}^d = 1$ can represent the dilation fields. When $\eta = 0.5$, and $\eta_o = 0.05$, the dilation field expands the boundary of the fluids to the boundary of their dilation fields by a distance approximately equal to half of the filter radius r . Take the 1D design problem shown in Fig. 5b as an example, both

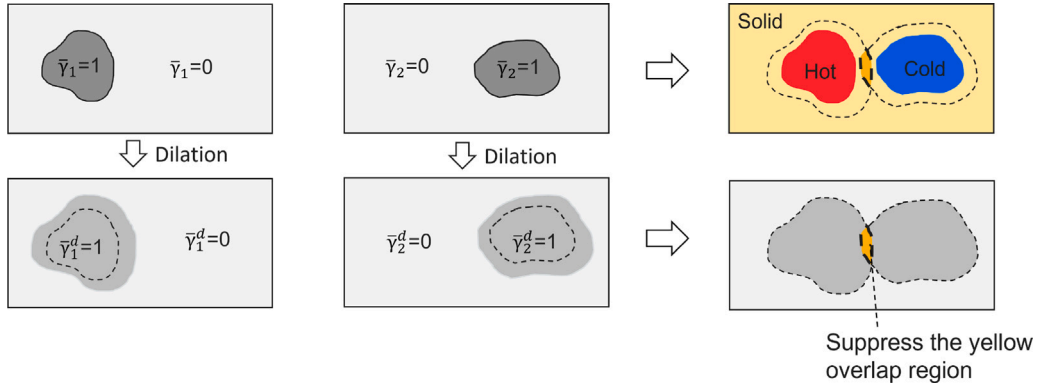
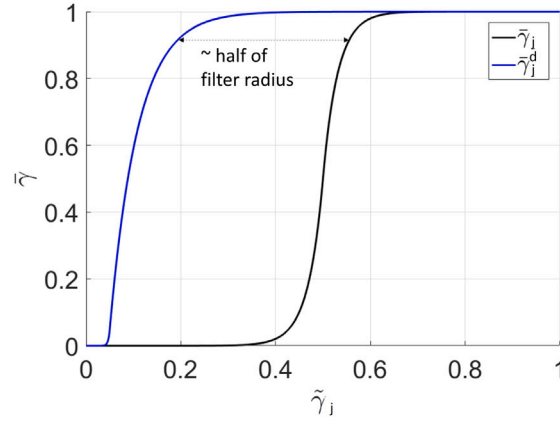
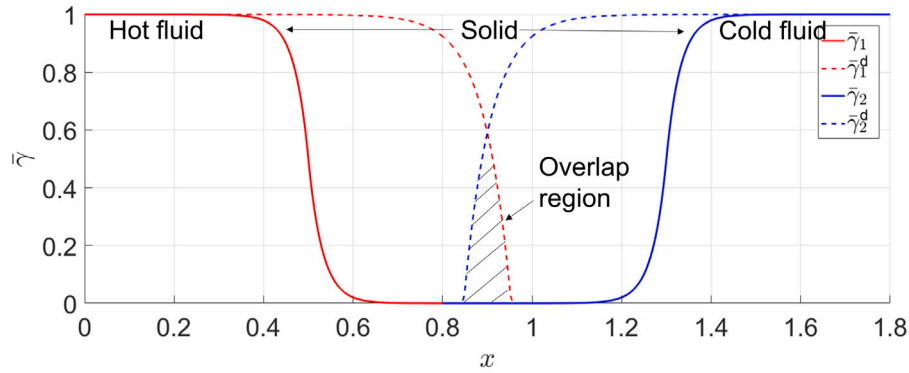


Fig. 4. The overlap region between two dilated density fields is used to control the minimal solid thickness. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



(a) Thickness of the dilation shell



(b) Minimal wall thickness

Fig. 5. Minimal thickness of the wall between the two fluids.

fluid boundaries are expanded to the boundaries of their dilation fields by a distance approximately equal to half of the filter radius r . The minimal wall thickness constraint (4) suppresses the overlapping of the positive region of the two dilation fields. These regions will be filled with $\bar{\gamma}_1 = 0$ and $\bar{\gamma}_2 = 0$, which represents the solid component. Consequently, Eq. (4) ensures a minimal solid wall thickness between the two fluids.

3.2.2. Density-gradient based overhang angle constraint

Removing support structures in metal additive manufacturing is particularly challenging, especially for internal structures. With no

tool access, the only way to remove support structures from internal cooling channels is through chemical polishing. This process extends the manufacturing steps and increases the final product's cost. Therefore, a self-supported heat exchanger is more desirable than one that requires support structures. Langelaar [26] presents self-supported topology optimized designs using an AM filter. This method involves updating the optimization field to generate a printable structure. In our study, we applied the method developed by Qian [20], which imposes constraints on the solid components. Eq. (7) represents the density-

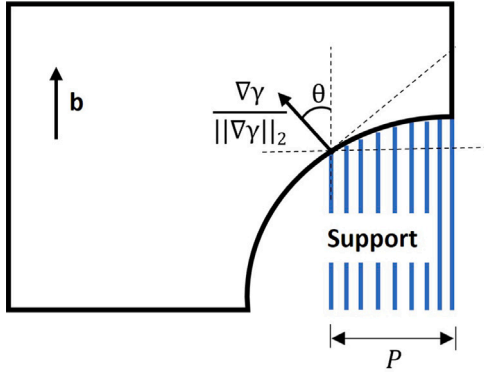


Fig. 6. Density-gradient based overhang angle constraint for ensuring self-supporting designs for additive manufacturing.

gradient based overhang angle constraint introduced by Qian [20]. With this constraint, the overhang angle of the solid component remains below the permissible angle throughout the optimization process. Consequently, the optimized structure is self-supported and can be printed without the need for support structures.

$$\int_{\Omega} \frac{1}{1 + e^{-2\beta_a(\mathbf{b} \cdot \tilde{\gamma}_s / \|\nabla \tilde{\gamma}_s\| - \cos(\bar{\theta}))}} \frac{\mathbf{b} \cdot \tilde{\gamma}_s}{\|\nabla \tilde{\gamma}_s\|} d\Omega \leq \bar{P}. \quad (7)$$

here, $\mathbf{b}$ signifies the build direction, and $\bar{\theta}$ denotes the maximum permissible overhang angle for self-supported structures. $\bar{\theta}$ is demonstrated in Fig. 6. The parameter β_a corresponds to the Heaviside parameter, it is equal to 16 in this study, and $\bar{P}$ is the threshold value of the constraint. The constraint is implemented by evaluating the left side of Eq. (7) to compute the projected undercut perimeter. The aggregate value of the overhang angle constraint is set to be no more than or equal to $\bar{P}$, a small numerical value. For the two-density method, we evaluate the solid phase in Eq. (7) as $\tilde{\gamma}_s = 1$, and $\tilde{\gamma}_s = (1 - \tilde{\gamma}_1)(1 - \tilde{\gamma}_2)$ which means $\tilde{\gamma}_s$ is the product of $1 - \tilde{\gamma}_1$ and $1 - \tilde{\gamma}_2$.

3.3. Topology optimization formulations

Our optimization problem focuses on enhancing the heat transfer rate of the two-fluid heat exchanger. The physics of the problem, such as velocity, pressure, and temperature are captured by solving governing equations. We utilize the Navier–Stokes (NS) equations to model the velocity and pressure fields and utilize the energy equation to obtain the temperature distribution.

3.3.1. Navier–Stokes equations

Two sets of Navier–Stokes equations for the hot and cold fluids are solved separately. These equations are used to determine the velocity and pressure drop for both fluids. For the hot fluid, we have:

$$-\nabla \cdot \nu_1 \nabla \mathbf{u}_1 + (\mathbf{u}_1 \cdot \nabla) \mathbf{u}_1 + \nabla p_1 = -\alpha_1(\tilde{\gamma}_1) \mathbf{u}_1 \quad \text{in } \Omega, \quad (8a)$$

$$-\nabla \cdot \mathbf{u}_1 = 0 \quad \text{in } \Omega, \quad (8b)$$

where ν_1 , $\mathbf{u}_1$ and p_1 are the kinematic viscosity, velocity and kinematic pressure of the hot fluid, respectively. $\alpha(\tilde{\gamma}_1)$ is the Brinkman friction term. The Rational Approximation of Material Properties (RAMP) method is used to calculate this term. The form is given in (11a).

Similar as the hot fluid, the following equations are solved for the velocity and pressure fields of the cold fluid:

$$-\nabla \cdot \nu_2 \nabla \mathbf{u}_2 + (\mathbf{u}_2 \cdot \nabla) \mathbf{u}_2 + \nabla p_2 = -\alpha_2(\tilde{\gamma}_2) \mathbf{u}_2 \quad \text{in } \Omega, \quad (9a)$$

$$-\nabla \cdot \mathbf{u}_2 = 0 \quad \text{in } \Omega, \quad (9b)$$

where ν_2 , $\mathbf{u}_2$ and p_2 are the kinematic viscosity, velocity and kinematic pressure of the cold fluid, respectively.

3.3.2. Energy equation

The energy equation is weakly coupled with the NS equations, as the velocity field $\mathbf{u}_1$ and $\mathbf{u}_2$ in the convection term of the energy equation are solved from equation ((8a)–(8b), (9a)–(9b)). We solve the temperature field from:

$$-\nabla \cdot (\kappa(\tilde{\gamma}_1, \tilde{\gamma}_2) \nabla T) + (\rho_1 C_{p1} \mathbf{u}_1 + \rho_2 C_{p2} \mathbf{u}_2) \cdot \nabla T = 0 \quad \text{in } \Omega, \quad (10)$$

where ρ_1 , C_{p1} , ρ_2 , and C_{p2} are the density and specific heat capacity of the hot fluid and the cold fluid, respectively. $\kappa(\tilde{\gamma}_1, \tilde{\gamma}_2)$ is the thermal conductivity interpolated between the hot fluid, the cold fluid, and the solid. The form is given in (11c).

3.3.3. Properties interpolation

Eqs. (11a)–(11c) show the RAMP form of the Brinkman friction term $\alpha_1(\gamma_1)$, $\alpha_2(\gamma_2)$, and bi-linear form of the thermal conductivity $\kappa(\gamma_1, \gamma_2)$.

$$\alpha_1(\tilde{\gamma}_1) = \alpha_{max1} \frac{q(1 - \tilde{\gamma}_1)}{q + \tilde{\gamma}_1}, \quad (11a)$$

$$\alpha_2(\tilde{\gamma}_2) = \alpha_{max2} \frac{q(1 - \tilde{\gamma}_2)}{q + \tilde{\gamma}_2}, \quad (11b)$$

$$\kappa(\tilde{\gamma}_1, \tilde{\gamma}_2) = \kappa_s(1 - \tilde{\gamma}_1)(1 - \tilde{\gamma}_2) + \kappa_{f1}\tilde{\gamma}_1 + \kappa_{f2}\tilde{\gamma}_2. \quad (11c)$$

Here, q is used to adjust the function curvature. With $\tilde{\gamma}_1 = 1$, which represents the hot fluid, $\alpha_1(1) = 0$. When $\tilde{\gamma}_1 = 0$, which represents the solid or the other fluid, $\alpha_1(0) = \alpha_{max1}$. A big value for α_{max1} means that the flow velocity of the hot fluid in the solid component or the cold fluid is low and close to 0. The same mechanism applies to (11b) for the NS equations of the cold flow. Thermal conductivity κ_s , κ_{f1} , and κ_{f2} in (11c) are the thermal conductivity of the solid component, the hot fluid, and the cold fluid, respectively. The bi-linear form is used to reduce the non-linearity of the optimization problem. When $(\tilde{\gamma}_1, \tilde{\gamma}_2) = (0, 0)$, which is the solid component, (11c) returns to the value of κ_s . Otherwise, when $\tilde{\gamma}_1 = 1$ or $\tilde{\gamma}_2 = 1$, (11c) returns to the thermal conductivity of the corresponding fluid.

3.4. Optimization problem

In the optimization problem, we iteratively update the two density fields γ_1 and γ_2 to evolve the design and maximize the heat transfer rate between the hot fluid and the cold fluid. In each optimization iteration, we solve the NS equations and energy equation in ((8a)–(8b), (9a)–(9b), (10)) for the velocity, pressure, and temperature. We calculate the objective function and constraints' value based on the following optimization problem.

$$\min_{\gamma_1, \gamma_2} J = \int_{\Gamma_{hout}} \rho_1 C_{p1} (T - T_{cold}) \mathbf{u}_1 \cdot \mathbf{n} ds + \int_{\Gamma_{cout}} \rho_2 C_{p2} (T_{hot} - T) \mathbf{u}_2 \cdot \mathbf{n} ds \quad (12a)$$

$$\text{s.t.} \quad \int_{\Gamma_{hin}} p_1 ds - \int_{\Gamma_{hout}} p_1 ds \leq P_1, \quad (12b)$$

$$\int_{\Gamma_{cin}} p_2 ds - \int_{\Gamma_{cout}} p_2 ds \leq P_2, \quad (12c)$$

$$(6), \quad (12d)$$

$$(7), \quad (12e)$$

$$\frac{\int_{\Gamma_{hout}} (\mathbf{u}_1 \cdot \mathbf{n} - U_1)^2 ds}{\pi(D/2)^2 U_1^2} \leq \Phi_1^2, \quad (12f)$$

$$\frac{\int_{\Gamma_{cout}} (\mathbf{u}_2 \cdot \mathbf{n} - U_2)^2 ds}{\pi(D/2)^2 U_2^2} \leq \Phi_2^2, \quad (12g)$$

Γ_{hout} , Γ_{hin} , Γ_{cout} , and Γ_{cin} are the outlet & inlet faces of the hot and cold fluids, respectively. T_{cold} and T_{hot} are the inlet temperatures of the cold fluid and hot fluids. The objective function (12a) is to maximize the heat transfer rate between the hot and cold fluids. (12b)–(12c) are the pressure drop constraints for the hot fluid and the cold fluid

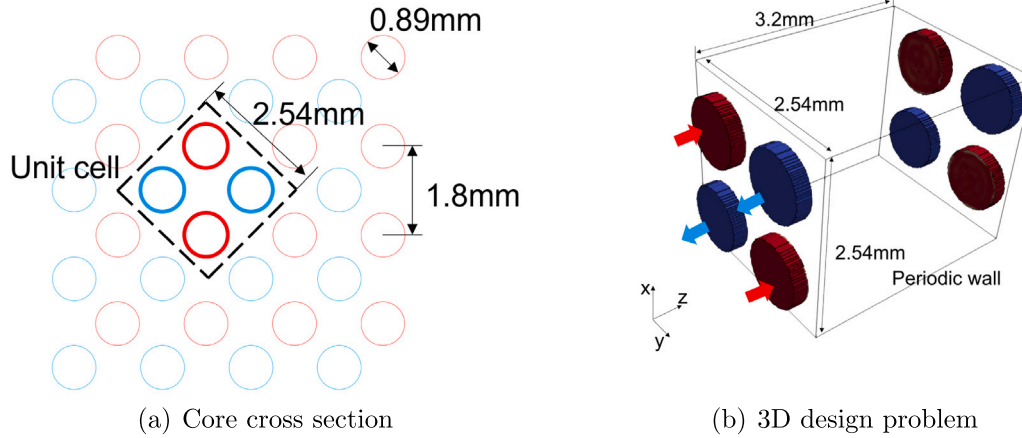


Fig. 7. Design specification of the unit cell. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

respectively. (12d)–(12e) are the two manufacturing constraints, the minimal thickness constraint and the overhang angle constraint. D is the diameter of the inlet and outlet extensions. U_1 and U_2 are the magnitudes of the inlet velocity of the hot and cold flow. (12f)–(12g) add a bound to the uniformity of the outlet velocity profile.

3.4.1. Optimization implementation

In the initial stage of the topology optimization, both γ_1 and γ_2 are set to 0.5. In each optimization iteration, the governing equations in ((8a)–(8b), (9a)–(9b), (10)) are solved. Subsequently, the continuous adjoint method is employed to compute the sensitivity of the cost and constraints with respect to the two density fields. Utilizing these sensitivities, the optimizer, Method of Moving Asymptotes (MMA) (Svanberg [27]), is utilized to update the density fields, aiming to minimize the cost while satisfying the constraints. The implementation of MMA utilized in this study follows the version developed by Aage and Lazarov [28]. Through this iterative process, the density fields evolve. The topology optimization process halts, and the design is extracted either when a maximum number of iterations is reached or when convergence is achieved.

In the numerical example presented below, a mesh comprising 6 million hexagonal cells is utilized to construct the unit cell heat exchanger. They are distributed on 512 CPUs. A total of 400 optimization iterations are conducted within 2 days. We implemented the solver of NS equation, energy equation, and the adjoint equations in open source CFD software OpenFOAM. The entire process is run on the Stampede2 supercomputer (Towns et al. [29]).

4. Numerical example

In this section, we demonstrate a numerical example of the topologically optimized heat exchanger. The design specification is shown in Fig. 7. A square cross section containing two hot and two cold channels was chosen as a unit cell. The diameter of the hot and cold channels is 0.89 mm. The distances between two adjacent hot channels and cold channels are both 1.8 mm. The 2D square cross section is shown in Fig. 7a. This is also the cross section of the benchmark straight channel heat exchanger. The length of the unit cell is 2.54 mm. The opposite sides are defined as periodic boundaries, implemented as cyclic boundary conditions in OpenFOAM. Table 1 outlines the boundary types used for the inlet, outlet, and periodic boundaries in OpenFOAM. The 3D design problem on the unit cell is shown in Fig. 7(b). The red and blue cylinder regions are flow inlet & outlet extensions. Each of them is 0.2 mm in length. The middle box with a size of 2.54 mm * 2.54 mm * 3.2 mm is the design domain. The four side walls of the design domain

Table 1

Boundary types used in OpenFOAM.

Boundary	Variable	OpenFOAM type
Inlet	Velocity	fixedValue
	Pressure	zeroGradient
	Temperature	fixedValue
Outlet	Velocity	zeroGradient
	Pressure	fixedValue
	Temperature	zeroGradient
Periodic boundary	Velocity	cyclic
	Pressure	cyclic
	Temperature	cyclic

Table 2

Material properties.

	$\nu(\text{m}^2/\text{s})$	$\rho(\text{kg}/\text{m}^3)$	$C_p(\text{J}/\text{kgK})$	$\kappa(\text{W}/\text{mK})$
Hot Fluid	$1.2\text{e}-6$	38.79	1266	0.081
Cold Fluid	$1.3\text{e}-7$	240.92	1285	0.05
Haynes 282				22.6

are periodic boundaries. The hot fluid enters from the left side and leaves from the right side. The cold fluid enters from the right side and leaves from the left side. Both the hot fluid and the cold fluid are supercritical CO_2 (sCO_2). The solid material is Haynes 282, a nickel-based superalloy renowned for its excellent high-temperature strength, resistance to oxidation, and creep properties. This material is chosen to design the heat exchanger to withstand high temperatures and high pressure. The material properties are assumed to be constant, as shown in Table 2.

The domain in Fig. 7(b) is decomposed into ~ 5.6 million hexagonal cells. The edge length of each cell is 0.02 mm. Based on the relationship between the filter radius and minimal wall thickness, mentioned in Section 3.2.1, we chose the filter radius to be 20 cell lengths. Thus, we can estimate the thickness of the solid wall of the optimized structure is $20 \times 0.02 = 0.4$ mm.

The hot fluid inlet temperature is 1053 K and the cold fluid inlet temperature is 573 K. Considering the other design parameters, and the boundary conditions in Table 3, the resulting topologically optimized hot fluid channel, cold fluid channel, and the solid component are shown in Fig. 8. The two hot channels merge into one, while the two cold fluid channels remain separate. Solid fins are observed at the interface between the fluid channels and the solid component. The convergence of the cost and constraints are shown in Fig. 9. The cost has been reduced while ensuring that all constraints are satisfied.

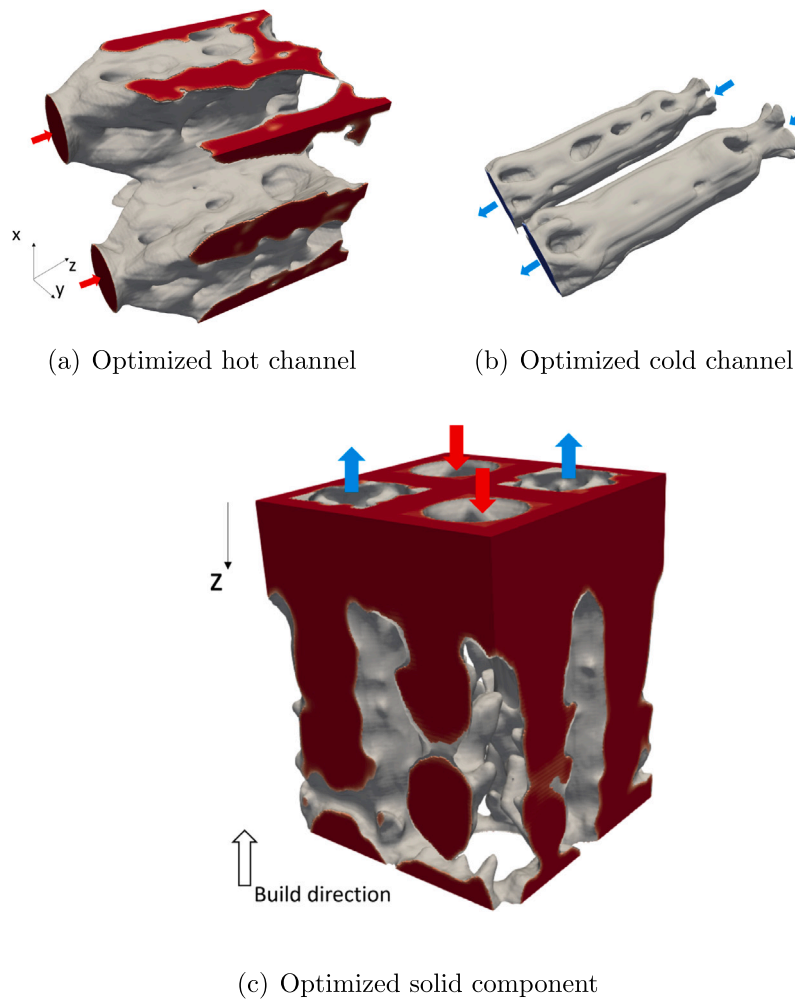


Fig. 8. Topologically optimized two-fluid heat exchanger with 45° overhang angle constraint.

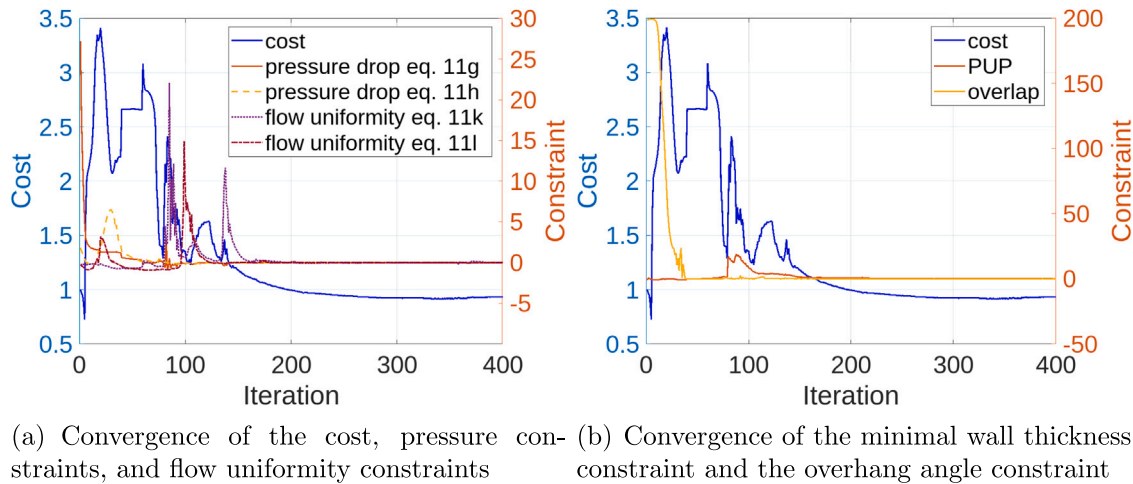


Fig. 9. Convergence of the cost and constraints. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 3
Optimization parameters.

U_1 (m/s)	U_2 (m/s)	P_1 (m ⁴ /s ²)	P_2 (m ⁴ /s ²)	Φ_o	$\bar{P}$	Φ_1	Φ_2
0.256	0.037	9.6e-6	1.56e-6	0.005	0.01	30%	30%

In this design, we chose the build direction $\mathbf{b}$ in the z direction or -z direction as shown in Fig. 7b, which means the heat exchanger will be printed in the same direction as the fluid flow. The choice of $\mathbf{b}$ would then be either (0, 0, 1) or (0, 0, -1).

In our numerical study, the optimized structure with $\mathbf{b} = (0, 0, 1)$ has a much higher cost value compared with the optimized structure

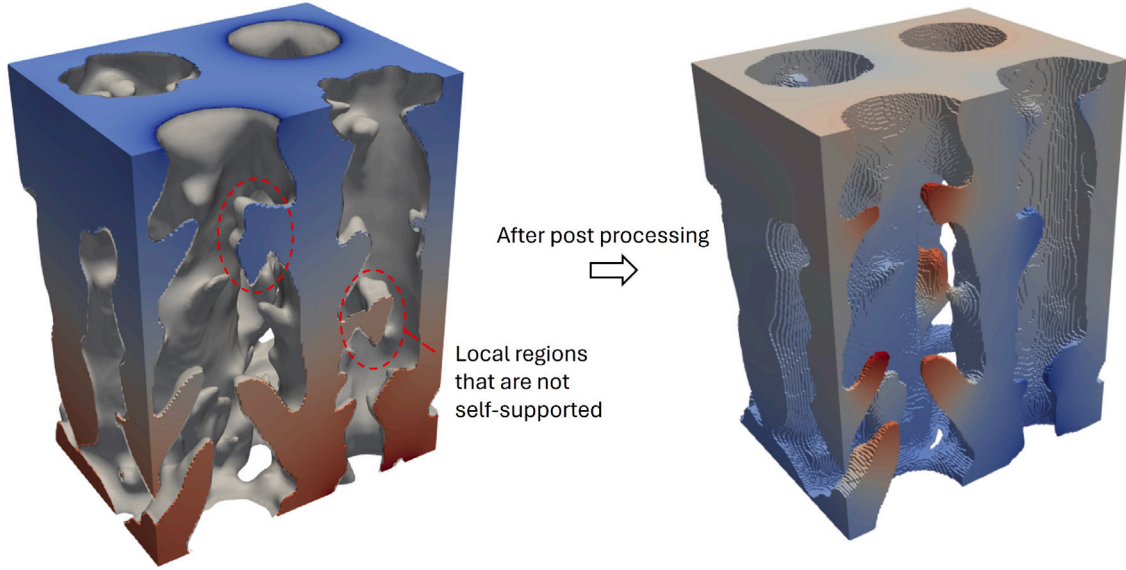


Fig. 10. Post processing of local regions that are not self-supported in the optimized design.

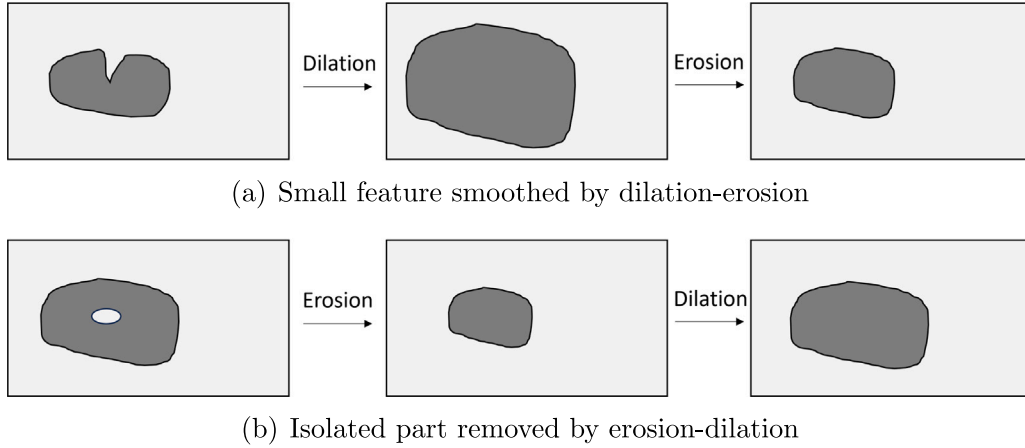


Fig. 11. Post processing with erosion and dilation.

with $\mathbf{b} = (0, 0, -1)$. The result of the TO design with a lower cost value is shown in Fig. 8. The overall structure satisfies the overhang angle constraint and can be self-supported. However, since (12e) is a volume integration form rather than a point-wise constraint, some local regions are not self-supported. Fig. 10 shows two of such regions. The structures are colored based on temperature fields to enhance visual contrast. After investigating the optimized structure, we found four local regions that cannot be self-supported. These regions were further modified with post processing.

4.1. Post processing

The post processing is conducted by using the erosion and dilation treatment on the optimized designs. A pair of erosion and dilation (or vice versa), as shown in Fig. 11, can be used to remove the small and isolated features. The local non-self-supported regions in the previous section can be removed with this post processing method.

We identified four local regions requiring modification. In these areas, we applied a large filter radius ($r = 20$ cell lengths) while deactivating the filter ($r = 0$) elsewhere. Consequently, the erosion-dilation and dilation-erosion processes were exclusively applied to the identified local regions. The erosion and dilation processes were imposed on the two density fields of the two fluid fields, $\bar{\gamma}_1$ and $\bar{\gamma}_2$ respectively. In the

erosion process, we chose $\eta_d = 0.95$ in Eq. (5). In the dilation process, we chose $\eta_d = 0.05$ in Eq. (5). Fig. 12 illustrates the outcome of the post-processing step, where the local regions that need to be modified have been smoothed and can now be self-supported, while the remaining regions have undergone minimal change.

4.2. CFD model validation

A 1D heat transfer model was developed and applied to validate the heat transfer rate of the straight channel heat exchanger. Fig. 13 illustrates the setup of the 1D heat transfer problem. The unit cell was discretized into N elements in the longitudinal direction. For the i th element, the following energy balance, and convective and conduction heat transfer relations were applied:

$$dQ_i = \dot{m}_H C_{p,H} (T_{H,i} - T_{H,i+1}), \quad (13a)$$

$$dQ_i = h_H A_w \left(\frac{T_{H,i} + T_{H,i+1}}{2} - T_{S,H,i} \right), \quad (13b)$$

$$dQ_i = \dot{m}_C C_{p,C} (T_{C,i} - T_{C,i+1}), \quad (13c)$$

$$dQ_i = h_C A_w \left(T_{S,C,i} - \frac{T_{C,i} + T_{C,i+1}}{2} \right), \quad (13d)$$

$$dQ_i = \frac{\kappa_S A_w}{t} (T_{S,H,i} - T_{S,C,i}), \quad (13e)$$

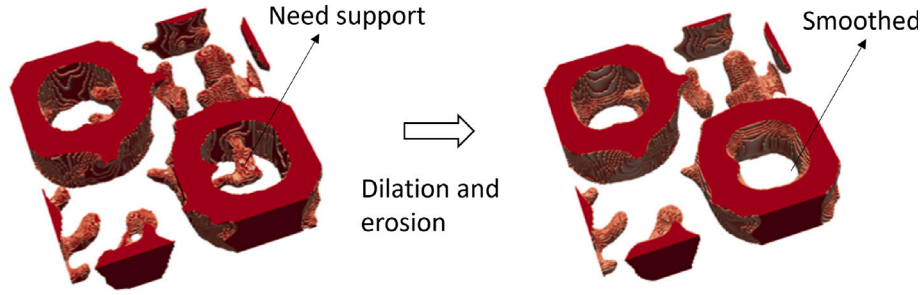


Fig. 12. Modified region becomes smooth and self-supported.

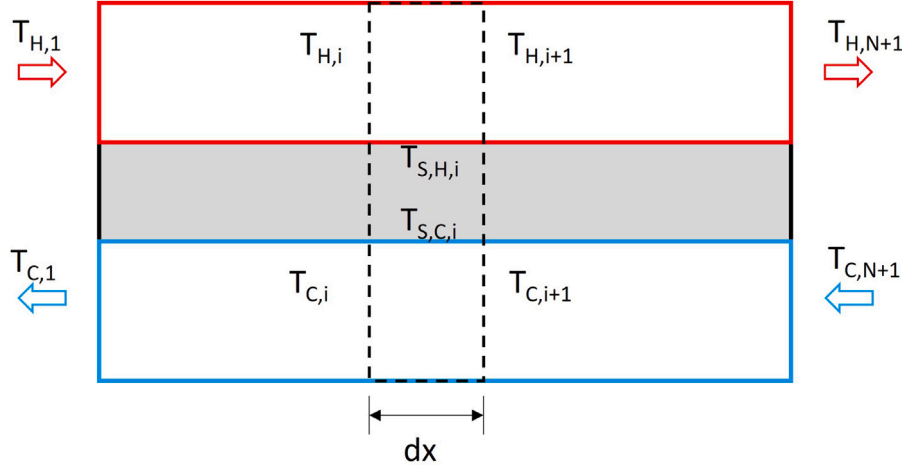


Fig. 13. 1D heat transfer model.

where dQ_i is the heat transfer rate in the i th element. $\dot{m}_H$ and $\dot{m}_C$ are the mass flow rates of the hot and cold fluids, respectively. $C_{p,H}$ and $C_{p,C}$ are the heat capacity of the hot and cold fluid shown in Table 2. $T_{H,i}$, $T_{C,i}$, $T_{H,i+1}$, and $T_{C,i+1}$ are the temperatures of the hot and cold fluids on either side of the element. $T_{S,H,i}$ and $T_{S,C,i}$ are the hot wall and cold wall temperatures in the element, assumed constant along dx . h_H and h_C are the local heat transfer coefficients of the hot and cold fluids. These coefficients are calculated as $h_H = \frac{\kappa_H Nu_H}{D}$, and $h_C = \frac{\kappa_C Nu_C}{D}$, where κ_H and κ_C are the thermal conductivities of the fluids, D is the channel diameter, and Nu_H and Nu_C are the Nusselt numbers. The Nusselt numbers are calculated based on Churchill and Ozoe [30] for laminar developing flow, equal to 4.36 for fully developed laminar flow, Gnielinski [31] for fully developed turbulent flow and Bhatti and Shah (Bhatti and Shah [32]) for developing turbulent flow. And $A_w = \pi D dx$ is the wetted area, and t is the wall thickness between the hot and cold channels.

To avoid any major issues (inaccuracies) associated with the flow development in very short channels, which would impact the accuracy of the methods, the CFD result and the 1D model was compared by simulating a long channel, as shown in Fig. 14, to allow for a fully developed flow to be established. The diameter and wall thickness are kept the same as the straight channel heat exchanger in the previous example, with the only change being the length, which was made 20 times longer, resulting in a length of 7.2 cm. The heat transfer rate based on our CFD simulation is 18.1 W.

Eqs. (13a)–(13e) were solved by Equation Engineering Solver (EES) (Nellis and Klein [33]) with $N = 100$. The boundary conditions were set as $T_{H,1} = 1073$ K, and $T_{C,N+1} = 573$ K. We obtained an outlet temperature of 836.1 K for the hot fluid and 811.1 K for the cold fluid, resulting in a heat transfer rate of 17.0 W. This value is very close to the heat transfer rate calculated from the CFD model, showing a deviation of only 5.9%. This error is within the accuracy of the convective

heat transfer coefficient prediction methods, and it is worthwhile to highlight that the conduction problem was very simplified in the 1D model, which impacts its accuracy. Therefore, the CFD model and simulation were considered validated.

4.3. Performance validation

We quantitatively assess the heat transfer rate as:

$$\begin{aligned} Q &= \int_{\Gamma_{hin}} \rho_1 C_{p1} T_{hot} \mathbf{u}_1 \cdot \mathbf{n} ds - \int_{\Gamma_{hout}} \rho_1 C_{p1} T_{u1} \cdot \mathbf{n} ds \\ &= \int_{\Gamma_{cout}} \rho_2 C_{p2} T_{u2} \cdot \mathbf{n} ds - \int_{\Gamma_{cin}} \rho_2 C_{p2} T_{cold} \mathbf{u}_2 \cdot \mathbf{n} ds, \end{aligned} \quad (14)$$

Additionally, the logarithmic average (LDMT) of the heat exchangers is computed as:

$$LMTD = \frac{(T_{hot,out} - T_{cold,in}) - (T_{hot,in} - T_{cold,out})}{\ln\left(\frac{T_{hot,out} - T_{cold,in}}{T_{hot,in} - T_{cold,out}}\right)}, \quad (15)$$

where $T_{cold,in}$, $T_{cold,out}$, $T_{hot,in}$ and $T_{hot,out}$ are the velocity-averaged cold inlet and outlet, and hot inlet and outlet mean temperature, respectively, as defined in Eq. (16).

$$T_{averaged} = \frac{\int_{\Gamma} \mathbf{u} \cdot \mathbf{n} T ds}{\int_{\Gamma} \mathbf{u} \cdot \mathbf{n} ds}, \quad (16)$$

here $T_{averaged}$ represents the velocity-averaged temperatures $T_{hot,out}$ and $T_{cold,out}$ on their corresponding faces, and Γ denotes the respective hot outlet, and cold outlet faces.

For performance validation, we applied the CFD model without the TO modification. This means that the Navier–Stokes equations were solved exclusively in the fluid regions, and only heat conduction was considered in the solid components.

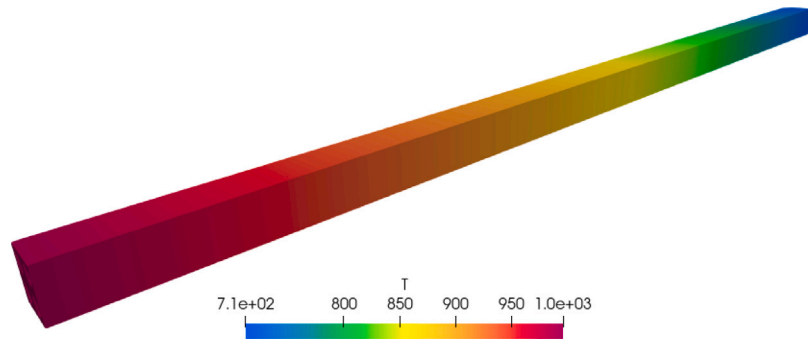


Fig. 14. 20 × length straight channel heat exchanger. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
Comparison of the straight channel and TO designs.

	Straight channel	TO design	Post-processing TO design
Heat Transfer Rate(W)	3.9	10.97	8.94
LMTD	3.13	5.39	5.09
Pressure	Hot fluid	136	103
Drop(Pa)	Cold fluid	143	122
Solid Volume(cm ³)	1.4e−2	1.19e−2	1.18e−2
Contact	Solid-Hot	0.20	0.57
Area(cm ²)	Solid-Cold	0.20	0.27

The post processing will influence the performance of the optimized heat exchanger. It is desirable to study how much the influence is and whether the final design after the modification can still perform better than the straight channel. Table 4 shows the performance of the topologically optimized design and the performance of the design after post processing. The straight channel heat exchanger in Table 4 has a diameter of 0.89 mm for both the hot and cold fluid channels. The geometry of the straight channel heat exchanger is shown in Fig. 17 (a). The hot fluid velocity is 1.28 m/s, and the cold fluid velocity is 0.185 m/s, which is 5 times the velocity used in Table 3. Other material properties are the same as Table 2.

From Table 4 we can see that the performance gain, measured as the heat transfer rate, of the TO design is $10.97/3.9 \approx 2.8$ times of the straight channel. Further modification reduces the heat transfer rate from 10.97 W to 8.94 W, reducing the gain over the straight channel by $8.94/3.9 \approx 2.3$ times. The optimized heat exchanger with manufacturability constraints still outperforms the straight channel in terms of heat transfer. Additionally, because the optimized design is lighter than the straight channel, it achieves an even higher power density. Power density, defined as the heat transfer rate divided by the weight of the heat exchanger, can be approximately 2.7 times greater for the post-processing topology-optimized (TO) design compared to the straight channel heat exchanger. This comparison is based on the following values: $8.94/3.9 \times 1.4/1.18 \approx 2.7$. Both heat exchangers are made of Haynes 282 and have the same solid density.

In Table 4, it is evident that the post-processing TO design approximately triples the contact area between the solid and hot fluid interface compared to the straight channel heat exchanger. However, it is essential to note that the increase in heat transfer rate cannot be solely attributed to the enlarged surface area. Instead, it stems from the improved cooling mechanism facilitated by the TO heat exchanger, which enhances convection by disrupting boundary layers.

Additionally, it is important to note the high pressure drop of the TO design. In Table 4, the pressure drop of the TO design is higher than that of the straight channel heat exchanger. This is because we set a loose pressure drop constraint to focus on optimizing the heat transfer rate.

In practice, the pressure drop of the straight channel heat exchanger is usually smaller than the allowable pressure drop. Thus, a loose pressure drop constraint is used in this study, leading to the higher pressure drop of the TO design compared to the straight channel.

4.4. Discussion

4.4.1. Cooling mechanism

To investigate the cooling mechanism of the optimized heat exchanger, we studied the thermal flow in the straight channel versus the TO design heat exchanger. Fig. 15 compares the thermal flow of the hot and cold fluids in the straight channel and in the post-processing TO design. In the straight channel heat exchanger, a thermal boundary layer forms on the fluid-solid interface, while the TO design disrupts this boundary layer, enhancing heat transfer by increasing the convection effect. These effects are more pronounced in Fig. 16, where we focus on the thermal flow of the hot fluid in the TO design. The color bar of the temperature is adjusted to the maximum and minimum temperatures of the hot fluid to show a strong contrast between the hot and cold temperatures. The solid fins in the downstream region come into contact with fluid at high temperatures, leading to the re-initialization of the thermal boundary layer. In Fig. 16b, once the hot stream is cooled down, it returns towards the inlet and recirculates within the hot fluid channel to enhance mixing with other hot streams, thereby promoting a strong convection effect. To further illustrate this statement, we maintained the diameter of the straight channel while increasing its length by factors of 2, 3, and 4 respectively. The flow rates of the hot fluid and cold fluid are also kept the same. This allowed us to study the heat transfer rate of the straight channel heat exchanger under varied lengths. In Fig. 17, it is shown that both the surface area and volume of the straight channel heat exchanger have increased proportionally with the change in length. The color highlights the solid component temperature. The blue and red ends indicate the low and high temperatures, respectively. Subsequently, we compared the performance of the straight channel heat exchanger at different lengths with that of the post-processing TO design in Table 5.

Table 5 illustrates that even with a fourfold increase in length and a higher surface area compared to the post-processing TO design, the straight channel heat exchanger fails to match the heat exchange capability of the post-processing TO design. Specifically, the power density of the fourfold straight channel heat exchanger is merely $\frac{7.99}{5.6e-2} \times \frac{1.18e-2}{8.94} = 18.8\%$ of that of the post-processing TO design. Despite this 4× straight channel heat exchanger achieving surface areas 1.4 times and 3.0 times larger than the post-processing TO design on the hot and cold fluid side respectively (totaling 1.9 times), the heat transfer rate of the post-processing TO design is $8.94/7.99 = 112\%$ of that of the straight channel heat exchanger.

4.4.2. Outlet flow uniformity

Eqs. (12f)–(12g) impose two flow uniformity constraints on the outlet velocity of the hot and cold fluid. With these constraints, good

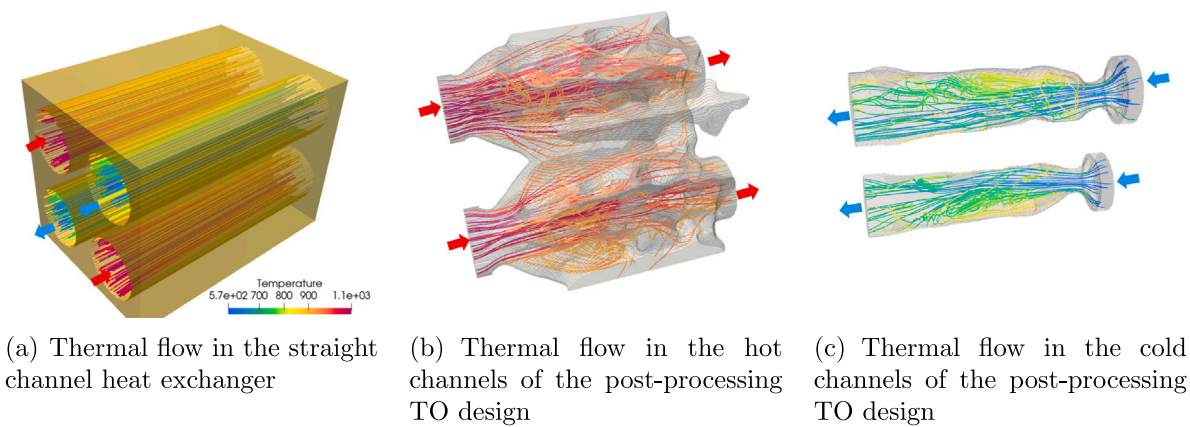


Fig. 15. Thermal flow in the straight channel heat exchanger and in the post-processing TO design. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

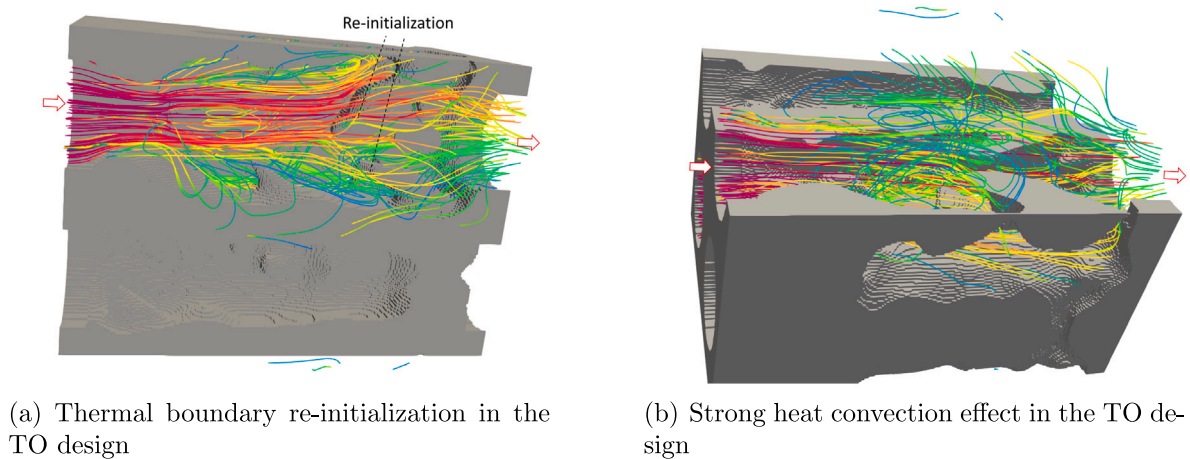


Fig. 16. Thermal flow of the hot fluid in the post-processing TO design.

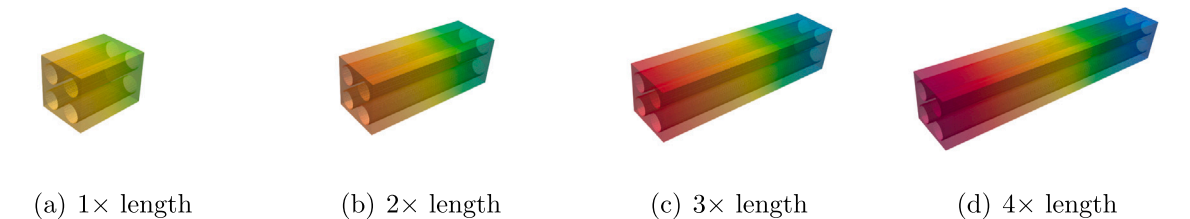
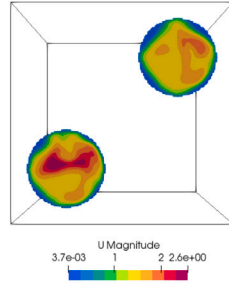


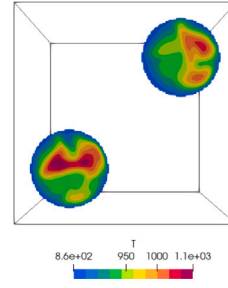
Fig. 17. Straight channel heat exchangers of different lengths. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Comparison of the straight channel at different lengths and TO designs.

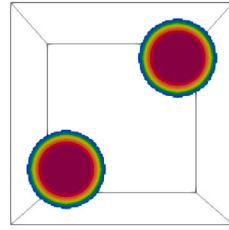
	Straight channel (original)	Straight channel (2×)	Straight channel (3×)	Straight channel (4×)	Post-processing TO design
Heat Transfer Rate(W)	3.9	5.55	6.86	7.99	8.94
LMTD	3.13	3.62	4.02	4.37	5.09
Pressure Hot fluid	32.7	47.6	60.1	71.3	103
Drop(Pa) Cold fluid	3.7	5.3	6.7	7.9	122
Solid Volume(cm ³)	1.4e−2	2.8e−2	4.2e−2	5.6e−2	1.18e−2
Contact Solid-Hot	0.20	0.4	0.6	0.8	0.57
Area(cm ²) Solid-Cold	0.20	0.4	0.6	0.8	0.27



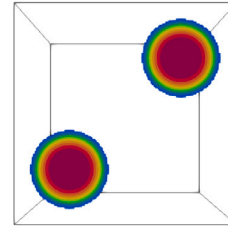
(a) post-processed TO design velocity



(b) post-processed TO design temperature



(c) Straight channel velocity



(d) Straight channel temperature

Fig. 18. Temperature and velocity on the hot fluid outlet surface from the CFD model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

periodicity can be expected for the unit cell design in the longitudinal direction. Fig. 18 compares the velocity and temperature of the hot fluid outlet between the TO design and the straight channel design. The velocity profile on the outlet surface of the TO design demonstrates overall good uniformity. However, since the result is from the CFD model without TO modification, there is a discrepancy in the velocity field compared to the velocity field from the TO model, which negatively affects outlet-velocity uniformity. Additionally, the geometry modification from the original TO design to the post-processed TO design also negatively impacts outlet-velocity uniformity. Furthermore, when comparing the hot fluid outlet temperature of the TO design with that of the straight channel heat exchanger, the TO design exhibits a well-mixed outlet temperature distribution, whereas the thermal boundary layer of the hot fluid adversely affects the heat transfer rate of the straight channel heat exchanger.

4.4.3. Solid thickness

Fig. 19 shows the fields with the two-density method on a cross section of the design. The cross section with a mid-plane cut is shown in Fig. 19 (a). The physical fields $\bar{\gamma}_1$ and $\bar{\gamma}_2$, which represent the hot fluid and cold fluid channels, are shown in Fig. 19 (b)& (c). The solid region is calculated with $(1 - \bar{\gamma}_1)(1 - \bar{\gamma}_2) = 1$. In Fig. 19 (d), a solid layer (in red) separates the two fluids (in blue). The solid layer is formed with the minimal wall thickness constraint described in (12d). With the dilation fields depicted in Fig. 19 (e)& (f). In Fig. 19 (g) shows the magnified black box of Fig. 19 (d). It can be seen that the solid layer formed between the two fluid channels has a thickness equal to 19 cell lengths. This thickness is close to the filter radius which is 20 cells wide. The overlap field, which is evaluated as $\bar{\gamma}_1^d \bar{\gamma}_2^d = 1$, in Fig. 19 (h) shows that the value of the overlap is small, which means the hot and cold fluid channels are successfully separated with the minimal wall thickness constraint.

Table 6

Longitudinal cross-section area and hydraulic diameter.

	Longitudinal cross-section area (mm ²)	Hot channel hydraulic diameter (mm)	Cold channel hydraulic diameter (mm)
Straight channel	3.96	0.89	0.89
Post-processed TO design	3.28	0.59	0.44

4.4.4. Longitudinal cross-section area and hydraulic diameter

The cross-sectional area and hydraulic diameter along the longitudinal direction provide crucial information for studying the heat exchanger design. We compared the straight channel design with the post-processed TO design. For the straight channel, the area and hydraulic diameter can be calculated from the channel diameter and solid thickness. For the post-processed TO design, we computed the average cross-sectional area of the solid and hydraulic diameter of the channels using the equations below:

$$S_s = \frac{V_s}{L}, \quad (17a)$$

$$D_H = \frac{4V_f}{S_f}, \quad (17b)$$

where S_s is the average longitudinal cross-section area of the solid, V_s is the volume of the solid, and L is the length of the unit cell heat exchanger in the longitudinal direction (z -direction in Fig. 7). D_H is the hydraulic diameter, V_f is the volume of the corresponding hot or cold channel, and S_f is the wetted surface area of the corresponding hot or cold channel. Table 6 shows the comparison between the straight channel and the post-processed TO design.

It can be seen from the table that the post-processed TO design has roughly half the hydraulic diameter compared to the straight channel.

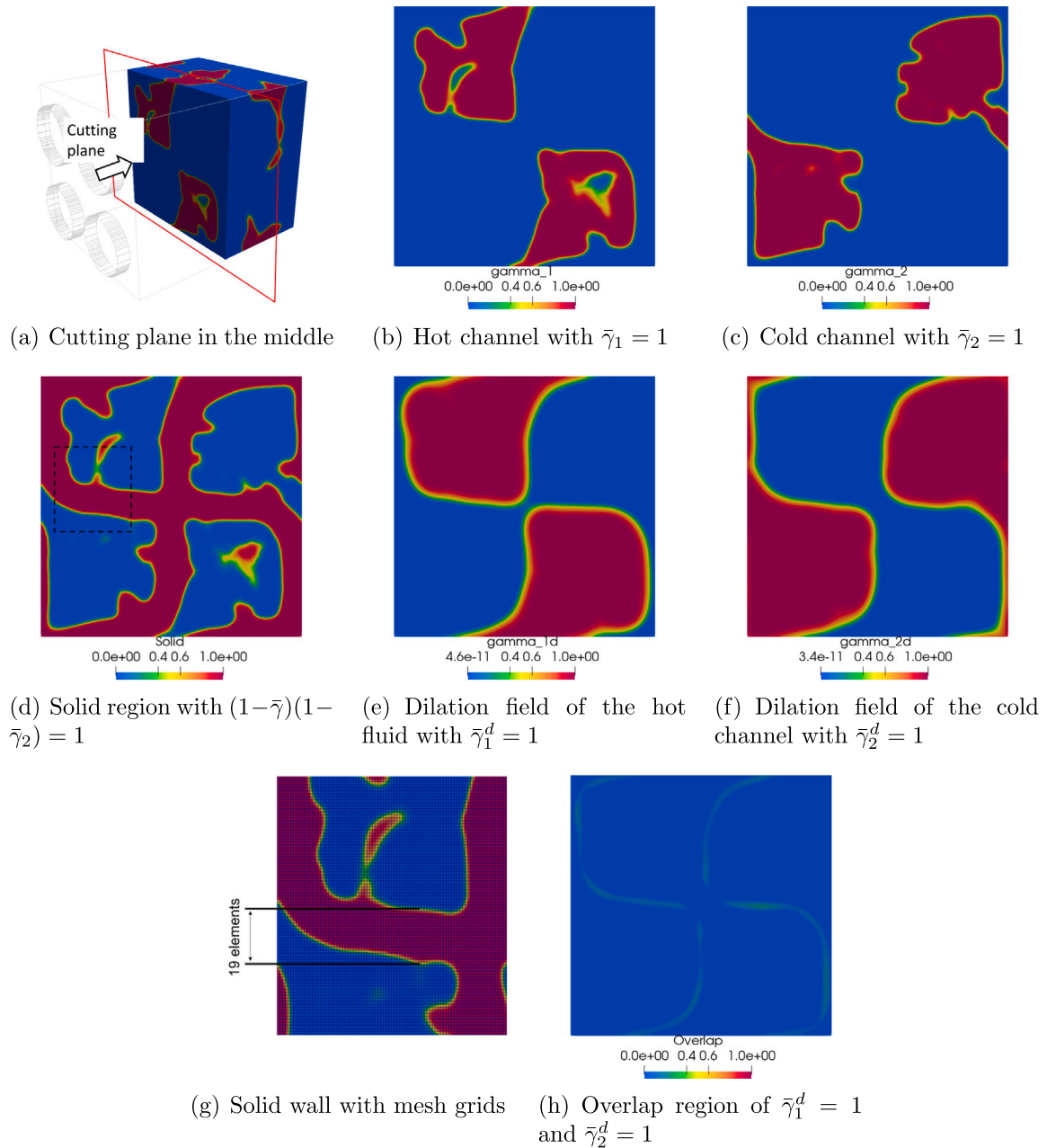


Fig. 19. Two density fields and the solid thickness. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

This also explains why the pressure drop of the TO design is higher than that of the straight channel, while the heat transfer is more efficient. A smaller hydraulic diameter typically results in a higher pressure drop and a higher heat transfer coefficient.

5. Additive manufacturing of the heat exchanger

The complete topologically optimized heat exchanger core is comprised of 512 topologically optimized unit cells. The heat exchanger core is shown in Fig. 20. A 0.5 mm thick wall is added on the four sides of the core. The total dimensions of the core are $21.3 \times 21.3 \times 28.8$ mm.

Laser Powder Bed Fusion (LPBF) is a subset of metal additive manufacturing capable of producing components with complex geometries and tight tolerances, making it the ideal technique for manufacturing the final heat exchanger. An EOS M290 (EOS GmbH, Krailling, Germany) system was used with Haynes 282 gas atomized powder (Höganäs AB, Sweden) with the particle size distribution of 15–40 μm

as feedstock. The system is equipped with a 400 W Ytterbium fiber laser, 1060-nm-wavelength, and 100- μm -diameter focused spot size. The full set of processing parameters for Haynes 282 was developed in-house and reported in Islam et al. [34]. In addition to the infill process parameters, which apply to the bulk of the heat exchanger, down and up skin parameter sets were used to improve the surface roughness of downward and upward facing surfaces, respectively. The thickness of each skin generated by the respective parameter set is approximately 120 μm . Moreover, a sophisticated contouring strategy was applied to further improve the roughness of resulting surfaces in the internal cooling channels of the heat exchanger. Since thin geometrical features, also known as edges, are prevalent in our design, it is critical to define laser scanning strategies suitable for these edges. The definitions of these parameters fall outside the scope of this work, but the values are reported here to promote the reproducibility of our results. Table 7 lists the main process parameters used to manufacture the heat exchangers. Additionally, the build plate was preheated to 80 $^{\circ}\text{C}$ and printing was

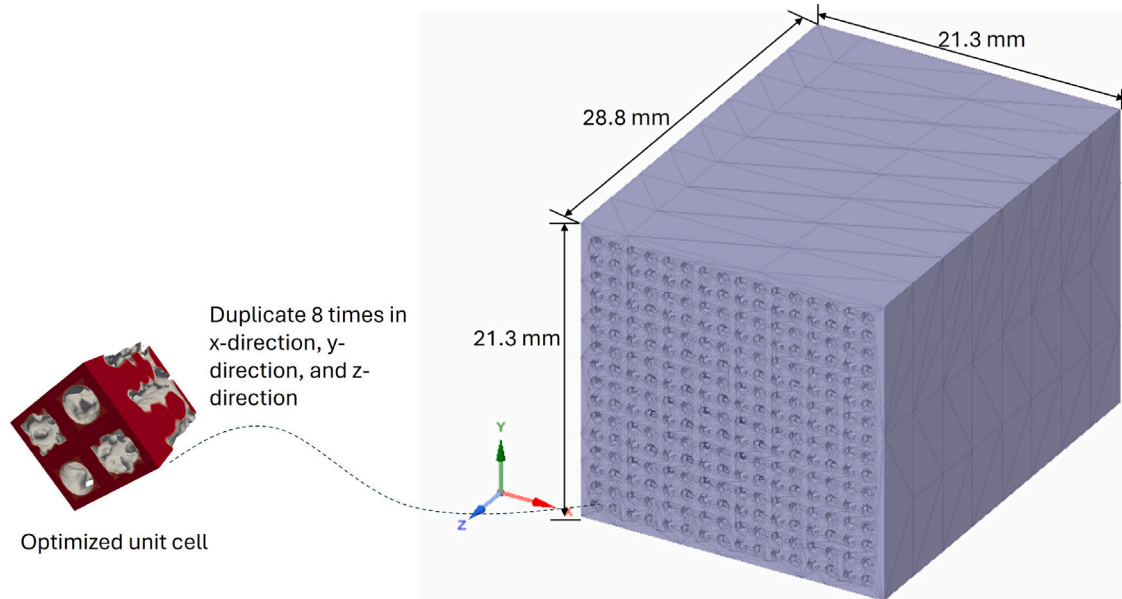


Fig. 20. The complete design model of the heat exchanger core with 512 optimized unit-cells.

Table 7
LPBF process parameters for Haynes 282.

	Infill	Up skin	Down skin	Contour 1	Contour 2	Edge
Power(W)	250	153	95	138	80	100
Velocity(mm/s)	980	600	785	390	800	900
Hatch spacing(mm)	0.11	0.99	0.11			

performed in an argon atmosphere. A 10-mm-wide scan stripe with a rotation angle of 67° was used at a constant layer thickness of 0.04 mm.

To demonstrate the robustness of our design and LPBF processing parameters, the heat exchangers were uniformly scaled in size to $2\times$ and $3\times$ larger. Additional cross sections of each heat exchanger were also printed to facilitate visual inspection of the topology optimized channels. The manufactured heat exchangers with their corresponding channels are shown in Fig. 21.

To delve deeper into the printing quality, X-ray scanning was employed to examine the internal structures. Fig. 22 depicts the X-ray scanning results of the original-sized printed heat exchanger core across various vertical cross-sections. Notably, the 45° self-supported fins are prominently displayed in the bottom-left figure. Furthermore, we conducted measurements of the wall thickness, as illustrated in Fig. 23. Upon observation, the wall thickness appears consistent across different locations. We selected a wall that visually appeared thinner than others for measurement. The chosen wall in the figure has a thickness of 0.45 mm, closely aligning with the minimal wall thickness constraint established during the optimization process. Visual inspection and X-ray analysis indicate successful manufacturing of the model and validate the chosen process parameters. Additively manufactured heat exchangers adequately captured the geometrical complexities of topology optimized models.

6. Conclusions

In this paper, we present a comprehensive approach to topology optimization for two-fluid heat exchangers, with a main focus on enhancing manufacturability and convective heat transfer rate. Three materials, the hot fluid, the cold fluid, and the solid are optimized simultaneously by using two density fields. Two sets of Navier–Stokes

equations are solved to model the hot and cold fluids. The heat exchanger is optimized to enhance the heat transfer rate. We consider two additive manufacturing requirements: the minimal thickness of the solid wall and the self-supported overhang angle. The minimal wall thickness is controlled with two parameters in the optimization process and the overhang angle of the design is controlled by applying the density-gradient based overhang angle constraint. Post-processing adjustments with erosion and dilation operations are made to the design to ensure its full manufacturability before its actual printing by an additive manufacturing process.

Our numerical examples showcase the efficacy of our approach, with optimized heat exchangers achieving remarkable improvements in heat transfer rates compared to traditional designs. Based on the CFD prediction, the optimized unit cell of the heat exchanger with imposed manufacturing constraints can have a 2.3-fold increase in heat transfer rate compared to the benchmark straight channel heat exchanger. The optimized heat exchanger improves the heat transfer rate by disrupting the thermal boundary layer and enhancing the thermal convection. When compared with a straight channel with 1.9 times higher surface area over the TO design, the TO design can still achieve a remarkable 5.3 times higher power density.

Moreover, the successful fabrication of the heat exchanger core consisting of topologically optimized unit cells further validates the manufacturability and practicality of our methodology. The printed heat exchangers exhibit excellent quality, with self-supported solid fins observed across all designs. Additionally, X-ray scanning was conducted to scrutinize the printed structures, confirming the presence of self-supported fins and revealing that the solid wall closely adheres to the minimal wall thickness constraint imposed during the optimization procedure. Overall, our study highlights the importance of integrating manufacturability considerations in the design process and demonstrates the feasibility of achieving optimized heat exchanger designs

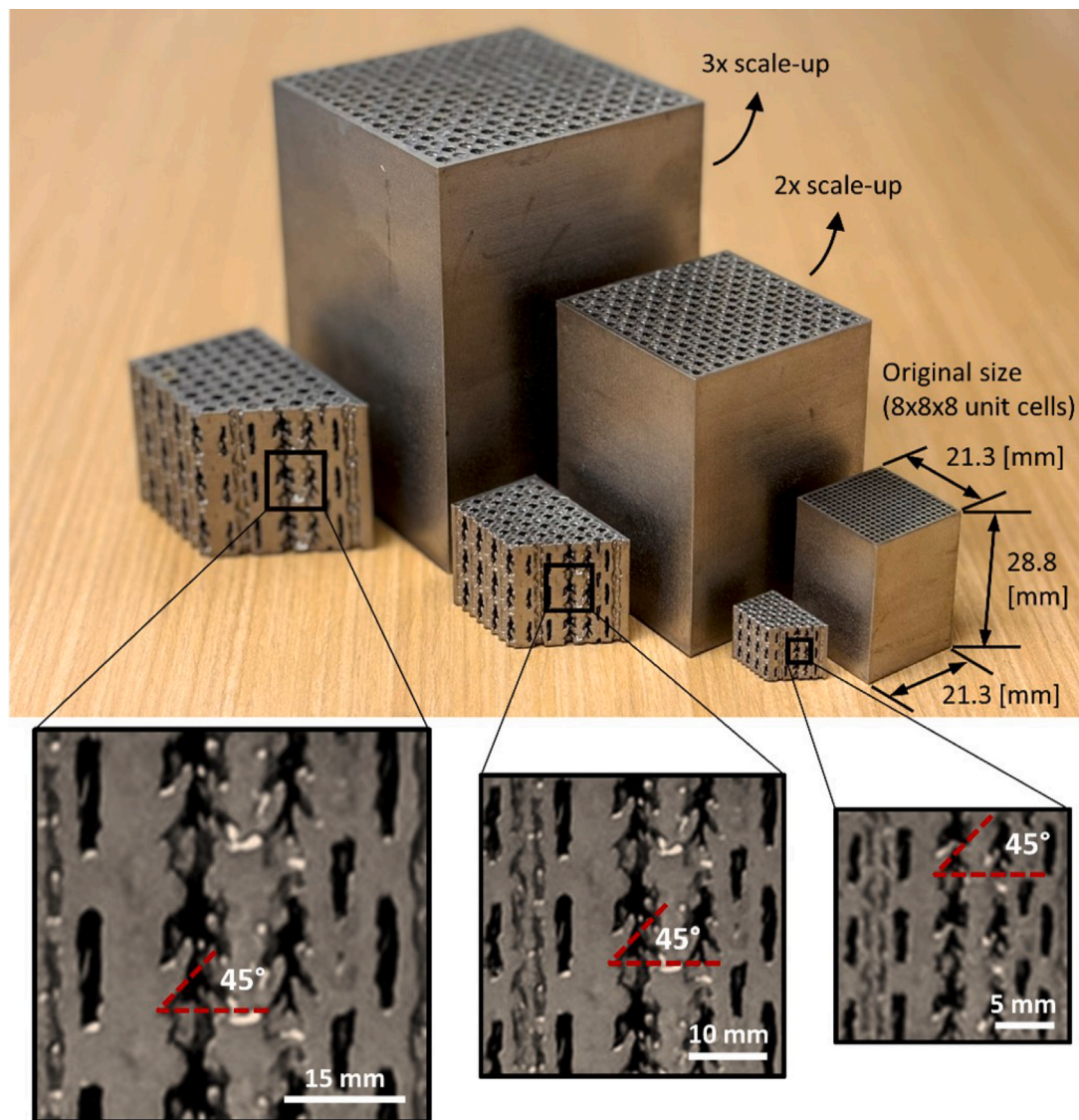


Fig. 21. Printed heat exchanger cores in three different sizes. The internal structures exhibit 45° self-supported fins.

suitable for additive manufacturing. We believe that our findings contribute to the advancement of topology optimization techniques in the field of heat exchangers and pave the way for the development of more efficient and manufacturable heat transfer systems.

Two significant challenges lie ahead for future work. Firstly, it is important to investigate the thermal stress characteristics of the optimized heat exchanger for high-temperature applications. Understanding and addressing potential thermal stresses is crucial for ensuring the structural integrity and longevity of the heat exchanger under high-temperature operating conditions. Secondly, conducting the thermohydraulic testing on the performance of the optimized heat exchanger, and comparing the results with the CFD prediction can help to further validate our work. This requires us to align the optimized heat exchanger with an efficient header. Addressing these challenges will further enhance the applicability of topology optimization in practical heat transfer applications.

CRediT authorship contribution statement

Sicheng Sun: Writing – original draft, Software, Methodology. **Behzad Rankouhi:** Writing – review & editing, Resources. **Dan J. Thoma:** Supervision. **Ian W. Jentz:** Visualization, Resources. **Mark H. Anderson:** Supervision. **Xiaoping Qian:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors of this paper declare that they have no financial or personal relationships that may have inappropriately influenced their work. Specifically, the authors have no conflicts of interest, financial or otherwise, that could be perceived as influencing the results or interpretation of the research reported in this paper.

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Data availability

Data will be made available on request.

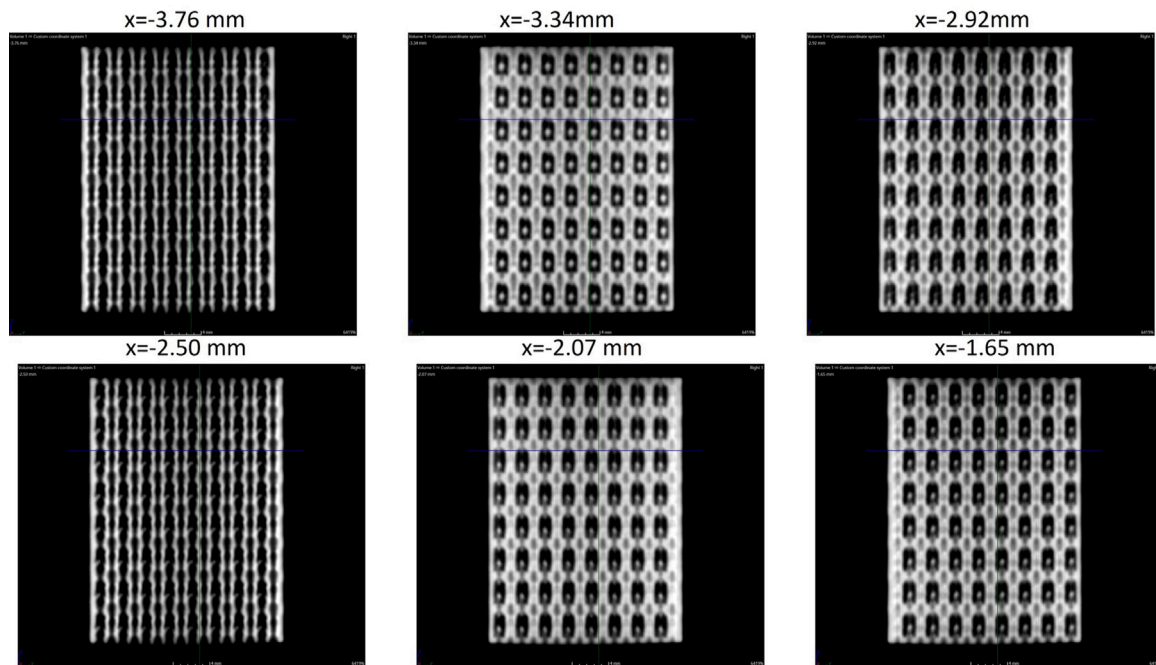


Fig. 22. X-ray scanning of the original-sized printed heat exchanger core at various vertical cross-sections.

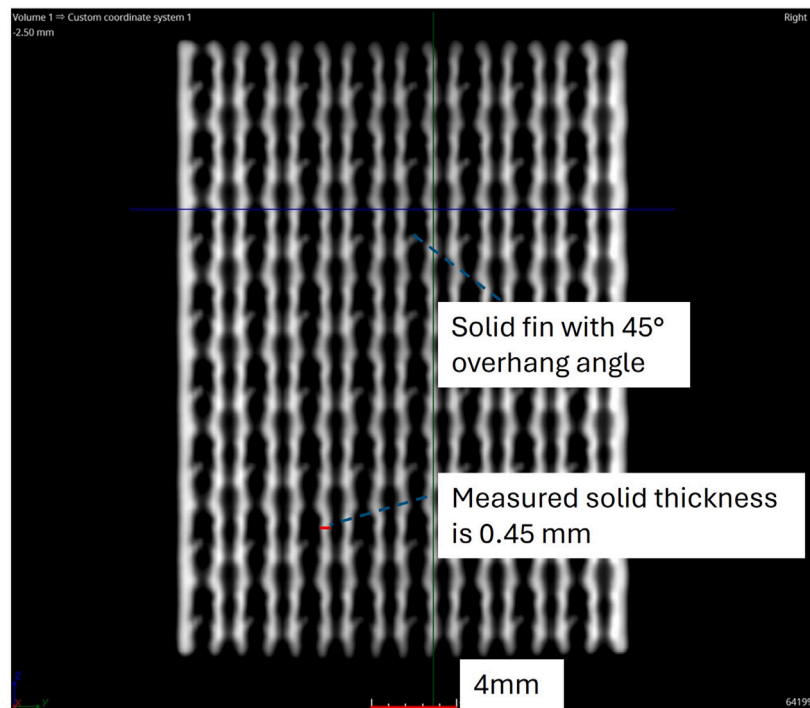


Fig. 23. Wall thickness measured in the picture is 0.45 mm.

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